# Akoizo Original Practice - Form 3

### Division B | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

**Name: _____ Date: _____**

**1. (2 points) Which word fits ABBC?**

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. MEET

B. TEAM

C. TREE

D. NOON

Answer: _____

**2. (2 points) What forces M = 1 in the ordinary SEND + MORE example?**

A. The extra leading carry

B. Spanish word frequencies

C. The letter M being common

D. Every M always means 1

Answer: _____

**3. (2 points) What is the key in a Caesar transformation?**

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

A. The recovered English sentence

B. The alphabet itself

C. The paper layout

D. The shift amount

Answer: _____

**4. (2 points) A fractionated-Morse triple can cross what?**

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. No boundary ever

B. Only complete words

C. An original letter boundary

D. Only a page boundary

Answer: _____

**5. (2 points) Which check is strongest after a plausible solve?**

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

A. Re-encrypt and compare every symbol

B. Accept any grammatical sentence

C. Ignore repeated letters

D. Count only words

Answer: _____

**6. (4 points) Explain “Polybius square” in your own words.**

Unit: Porta, Nihilist, and checkerboard tables.
Worked-case data: Nihilist ciphertext: 65. Key coordinate: 23. Square entry: Row 4, column 2 contains R.
Separate investigation observations:
Nihilist ciphertext: 65.
Key coordinate: 23.
Square entry: Row 4, column 2 contains R.

Answer: _____
_____
_____

**7. (2 points) Which priority follows the example assumptions?**

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

A. A before B

B. B necessarily before A

C. Both guarantee full credit

D. Neither can be compared

Answer: _____

**8. (6 points) Solve this case and explain your reasoning: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.**

Answer: _____
_____
_____

**9. (4 points) Encode C in the modern Baconian table.**

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

Answer: _____
_____
_____

**10. (6 points) Solve this case and explain your reasoning: Rectangle: Six letters in two columns. Ciphertext pieces: CTO and ATO. Clue: The plaintext starts with CAT.**

Answer: _____
_____
_____

**11. (4 points) Why cannot a triple always be read as one Morse letter?**

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

Answer: _____
_____
_____

**12. (4 points) A guessed word works once but contradicts another occurrence. What should change?**

Answer: _____
_____
_____

**13. (2 points) Define “Polybius square” in the context of Codebusters.**

Answer: _____
_____
_____

**14. (4 points) Give a prevention rule for wrong key direction.**

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

Answer: _____
_____
_____

**15. (2 points) Define “Atbash” in the context of Codebusters.**

Answer: _____
_____
_____

**16. (4 points) Describe a complete verification.**

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

Answer: _____
_____
_____

**17. (2 points) Define “Column height” in the context of Codebusters.**

Answer: _____
_____
_____

**18. (6 points) Solve this case and explain your reasoning: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?**

Answer: _____
_____
_____